

ISO14443A Air Interface Protocol Library function
V1.2

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1. Instruction

The development kit supports the ANSI-C, ANSI-C++ and other languages to call the C function interface, such as delphi, VB, C#. The development kit can not be used independently and must be combined with other RFID development kit. Before using, you should read "DLL Development Guide.doc" first, to make sure well understand the whole development process for SDK package.

2. Application Integration

This package consists of the following files:

File	Description
rfidlib_aip_iso14443A.dll	Dynamic link library, contains all API interface
rfidlib_aip_iso14443A.lib	VC++ project, static link library
rfidlib_aip_iso14443A.h	Header file of C/C++ project , the function definition
rfidlib_aip_iso14443A.cs	C# class file
rfidlib_aip_iso14443A.pas	Header file of Delphi function definition

Programming Language	Description
VC++	Copy rfidlib_aip_iso14443A.lib and rfidlib_aip_iso14443A.h to the project directory and add to the project. Copy rfidlib_aip_iso14443A.dll to exe file output directory.
C#	Copy rfidlib_aip_iso14443A.cs to the project directory and add to the project. Copy rfidlib_aip_iso14443A.dll to the exe file output directory.
Delphi	Copy rfidlib_aip_iso14443A.pas to the project directory and add to the project. Copy rfidlib_aip_iso14443A.dll to exe file output directory.

3. Interface of API

3.1. Overview

The interface of API is developed based on rfidlib_reader.dll, and defines standard protocol interface of ISO14443, which is convenient for application developers to integrate the ISO14443A protocol.

3.2. ISO14443A_GetLibVersion

Definition	DWORD ISO14443A_GetLibVersion(LPTSTR buf , DWORD nSize)		
Instruction	Get the version number of dynamic library		
Parameter	Name	Type	Remarks
	buf	LPTSTR	The string buffer used for saving the version number.
	nSize	DWORD	The maximum savable string in buffer
Return (DWORD)	Returns the number of characters written to the buffer, if the return value is equal to "nSize", shows that the version number length is bigger than the max number of characters of the buffer.		
Reference Code	<pre>char vbuf[128] ; DWORD written ; written = get_version(vbuf,128) ; if(written > 0) { /* show version string here */ printf(vbuf) ; } if(written == 128) { /* overflow,need to increase vbuf */ }</pre>		

3.3.ISO14443A_CreateInvenParam

Definition	RFID_DN_HANDLE ISO14443A_CreateInvenParam(RFID_DN_HANDLE hInvenParamSpecList, BYTE AntennaID)			
Instruction	Create ISO14443A Inventory Parameter data nodes for “RDR_TagInventory”.			
Parameter	Name	Type	Direction	Remarks
	hInvenParamSpecList	RFID_DN_HANDLE	[in]	The handle of Inventory parameter list created by RDR_CreateInvenParamSpecList.
	AntennaID	BYTE	[in]	The antenna ID , 0 means all of the antenna interface will read the tags which supported ISO14443A protocol. Other values for the corresponding antenna interface.
Return (RFID_DN_HANDLE)	If success, returns the handle of data node.			
Reference Code				

3.4.ISO14443A_ParseTagDataReport

Definition	err_t ISO14443A_ParseTagDataReport(RFID_DN_HANDLE hTagReport, DWORD *aip_id, DWORD *tag_id, DWORD * ant_id, BYTE uid[], BYTE *uidlen)			
Instruction	Parse ISO14443A tag data information. Obtained data node from RDR_GetTagDataReport, for the RDR_TagInventory function.			
Parameter	Name	Type	Direction	Remarks
	hTagReport	RFID_DN_HANDLE	[in]	The node handle of the tag data information
	aip_id	DWORD *	[out]	Return to the air interface protocol type ID, please refer to the air interface protocol type ID list of DLL Development Guide.doc.
	tag_id	DWORD *	[out]	Return to the tag types ID , refer to the tag types ID list of DLL Development Guide.doc .
	ant_id	DWORD *	[out]	Return antenna interface ID.
	uid	BYTE []	[out]	Return serial number of ISO14443A tag information
	uidlen	BYTE *	[in] [out]	[in] represents the maximum allowed length of UID to be passed in [out] represents the actual obtained UID length
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference code				

3.5.ISO14443A_Connect

Definition	err_t RFIDLIB_API ISO14443A_Connect(RFID_READER_HANDLE hr, BYTE tagType, BYTE uid[], BYTE uidLen, RFID_TAG_HANDLE *ht);			
Instruction	ISO14443A universal connection interface			
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader operation handle
	tagType	DWORD	[in]	The tag type ID , refer to the tag types ID list of “DLL Development Guide.doc”.
	uid	Byte[]	[in]	The tag uid data
	uidLen	BYTE	[in]	uid length
	ht	RFID_TAG_HANDLE *	[out]	Tag operation handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference code				

3.6.ISO14443Ap4_Transceive

Definition	err_t RFIDLIB_API ISO14443Ap4_Transceive(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE sndData[], DWORD sndLen, BYTE rcvBuff[], DWORD *nSize);			
Instruction	ISO14443A tag transparent transmission command			
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	sndData	Byte[]	[in]	Data sent to the tag
	sndLen	BYTE	[in]	The length of the data sent to the tag
	rcvBuff	BYTE	[out]	Received data buffer
	nSize	DWORD *	[in] [out]	[in] indicates the maximum length of data that can be received. [out] indicates the length of the actual received data
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference code				

3.7.MIFARE® classic (S50 and S70)function interface

3.7.1.MFCL_Connect 【Connect tag】

Definition	err_t MFCL_Connect(RFID_READER_HANDLE hr,BYTE tagType,BYTE uid[] /* 4 bytes */,RFID_TAG_HANDLE *ht)				
Instruction	Activate mifare classic tag				
Parameter	Name	Type	Direction	Remarks	
	hr	RFID_READER_HANDLE	[in]	Reader handle	
	tagType	BYTE	[in]	Tag Type:	
				Value	Direction
				0	MIFARE®S50
	1	MIFARE®S70			
	uid	BYTE[]	[in]	Tag serial number	
ht	RFID_TAG_HANDLE *	[out]	If successful, the tag handle output .		
Return (err_t)	0: Success Not 0: Failure, please check Error code table.				
Reference code					

3.7.2.MFCL_ReadBlock 【Read block】

Definition	err_t MFCL_ReadBlock(RFID_READER_HANDLE hr ,RFID_TAG_HANDLE ht, BYTE blkAddr ,BYTE blkData[], DWORD nSize) ;			
Instruction	Read block			
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	blkAddr	BYTE	[in]	Data block Address
	blkData	BYTE[]	[out]	The byte buffer is used to hold the data block has read.
	nSize	DWORD	[in]	The maximum number of bytes in the buffer .
Return (err_t)	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.7.3.MFCL_WriteBlock 【Write block】

Definition	err_t MFCL_WriteBlock(RFID_READER_HANDLE hr,RFID_TAG_HANDLE ht, BYTE blkAddr ,BYTE blkData[] /* 16 bytes */);			
Instruction	Write block			
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	blkAddr	BYTE	[in]	Block Address
	blkData	BYTE[]	[in]	Prepare the 16-byte block data to be written
Return (err_t)	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.7.4. MFCL_Authenticate 【Authentication】

Definition	err_t MFCL_Authenticate(RFID_READER_HANDLE hr,RFID_TAG_HANDLE ht, BYTE blkAddr ,BYTE keyType,BYTE key[]) ;			
Instruction	Key Authentication			
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	blkAddr	BYTE	[in]	Block Address
	keyType	BYTE	[in]	MIFARE® classic Authentication key type
				ValueDescription
				0Key A
				1Key B
	key	BYTE[]	[in]	6 bytes of key data
Return (err_t)	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.7.5. MFCL_Increment 【Increment】

Definition	err_t MFCL_Increment(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE blkAddr ,DWORD val)			
Instruction	Recharge, before using recharge, the data block must first be formatted with the value block by MFCL_FormatValueBlock.			
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	blkAddr	BYTE	[in]	Data block Address
	val	DWORD	[in]	Recharge Quota
Return (err_t)	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.7.6. MFCL_Decrement 【Decrement】

Definition	err_t MFCL_Decrement(RFID_READER_HANDLE hr,RFID_TAG_HANDLE ht, BYTE blkAddr ,DWORD val) ;			
Instruction	Decrement, before using decrement, the data block must first be formatted with the value block by MFCL_FormatValueBlock.			
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	blkAddr	BYTE	[in]	Data block address
	val	DWORD	[in]	Decrement Quota
Return (err_t)	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.7.7. MFCL_Restore 【Backup】

Definition	err_t MFCL_Restore(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE blkAddr) ;			
Instruction	Backup amount, before using backup, the data block must first be formatted with the value block by MFCL_FormatValueBlock.			
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	blkAddr	BYTE	[in]	Block address
Return (err_t)	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.7.8. MFCL_FormatValueBlock 【FormatValueBlock】

Definition	err_t MFCL_FormatValueBlock(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht,BYTE blkAddr,DWORD initValue) ;			
Instruction	Format the value data block			
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	blkAddr	BYTE	[in]	Block address
	initValue	DWORD	[in]	Initialization Value
Return (err_t)	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.7.9. MFCL_CreateAccessCondition 【CreateAccessCondition】

Definition	err_t MFCL_CreateAccessCondition(BYTE blk0AccType, BYTE blk1AccType,BYTE blk2AccType,BYTE trailerAccType, BYTE formattedAccCondi[]);			
Instruction	Create sector access condition			
Parameter	Name	Type	Direct ion	Remarks
	blk0AccType	BYTE	[in]	The access condition type of the sector data block 0, please refer to “Appendix 2. Data block access conditions of MIFARE® classic”
	blk1AccType	BYTE	[in]	The access condition type of the sector data block 1, please refer to “Appendix 2. Data block access conditions of MIFARE® classic”
	blk2AccType	BYTE	[in]	The access condition type of the sector data block 2, please refer to “Appendix 2. Data block access conditions of MIFARE® classic”
	trailerAccType	BYTE	[in]	The access condition type of the sector tail block, please refer to “Appendix 3. Tail block access conditions of MIFARE® classic”
	formattedAccCon di	BYTE[]	[out]	Output formatted 4 bytes can be written to the 6,7,8,9 bytes of the sector tail block.
Return (err_t)	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.7.10. MFCL_Sector2Block 【Sector Number convert to the block number】

Definition	BYTE MFCL_Sector2Block(BYTE sector)			
Description	Convert sector address to block address			
Parameters	Name	Type	Direction	Remarks
	sector	BYTE	[in]	Sector address
Return (BYTE)	Data block address			
Reference code				

3.7.11. MFCL_Block2Sector 【 Block number convert to the sector number】

Definition	BYTE MFCL_Block2Sector(BYTE block) ;			
Description	Convert data block address to sector address			
Parameters	Name	Type	Direction	Remarks
	block	BYTE	[in]	Data block address
Return (BYTE)	Sector address			
Reference code				

3.7.12. MFCL_IsTailerBlock 【Judge SectorTail Block】

Definition	BOOLEAN MFCL_IsTailerBlock(BYTE blkAddr)			
Description	Judged this data block whether is the sector tail block			
Parameters	Name	Type	Direction	Remarks
	blkAddr	BYTE	[in]	Data block address
Return (BOOLEAN)	Value	Description		
	0	Not sector tail block		
	1	It is sector tail block, including Key A , Key B, etc.		
Reference code				

3.7.13. MFCL_ParseAccessCondi 【Parse access condition】

Definition	err_t MFCL_ParseAccessCondi(BYTE formattedAccCondi[], BYTE *blk0AccType, BYTE *blk1AccType, BYTE *blk2AccType, BYTE *trailerAccType)			
Description	Parse sector access condition			
Parameters	Name	Type	Direction	Remarks
	formattedAccCondi	BYTE[]	[in]	The sixth, seventh, eighth, and ninth bytes of the tail block of the incoming sector
	blk0AccType	BYTE*	[out]	The access condition type of the data block 0 of the output sector
	blk1AccType	BYTE*	[out]	The access condition type of the data block 1 of the output sector
	blk2AccType	BYTE*	[out]	The access condition type of the data block 2 of the output sector
	trailerAccType	BYTE*	[out]	The access condition type of the tail block of the output sector
Return (err_t)	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.8.MIFARE ® ultralight Function interface

3.8.1.ULTRALIGHT_Connect 【Connect tag】

Definition	err_t ULTRALIGHT_Connect(RFID_READER_HANDLE hr,BYTE uid[] /* 7 bytes */ ,RFID_TAG_HANDLE *ht)			
Description	Connect Ultralight tag			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	uid	BYTE[]	[in]	7-byte serial number
	ht	RFID_TAG_HANDLE *	[out]	If successful,output tag handle
Return	0: Success Not 0: Failure , please check Error code table.			
Reference code				

3.8.2. ULTRALIGHT_Read 【Read data page】

Definition	err_t ULTRALIGHT_Read(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE pageNo , BYTE pagebuf[] , DWORD *nSize) ;			
Description	Read data page, from the specified page address continuous 4 pages total 16 bytes return.			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ULTRALIGHT_connect
	pageNo	BYTE	[in]	Page address0-15
	pagebuf	BYTE []	[out]	The buffer is used to hold the page data. The buffer size need to > = 16.
	nSize	DWORD *	[in,out]	In: The byte size of the buffer. Out: The number of bytes written to the buffer.
Return	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.8.3. ULTRALIGHT_Write 【Write data page】

Definition	err_t ULTRALIGHT_Write(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE pageNo, BYTE pagedata[])			
Description	Write data page			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	The reader drive handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle,created by ULTRALIGHT_connect .
	pageNo	BYTE	[in]	Page address3-15, Page 3 is the OTP page, and page 4-15 is the user page area.
	pagedata	BYTE[]	[in]	4 bytes of data
Return	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.8.4.ULTRALIGHT_LockPage 【Lock page】

Definition	err_t ULTRALIGHT_LockPage(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE pageType) ;																																																			
Description	Lock page, once the data page is locked, you can not change the data.																																																			
Parameters	Name	Type	Direction	Remarks																																																
	hr	RFID_READER_HANDLE	[in]	Reader handle																																																
	ht	RFID_TAG_HANDLE	[in]	The tag handle created by ULTRALIGHT_connect																																																
	pageType	BYTE	[in]	Lock page type																																																
				<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Lock page 3 (OTP Area)</td></tr><tr><td>1</td><td>Lock page 4</td></tr><tr><td>2</td><td>Lock page 5</td></tr><tr><td>3</td><td>Lock page 6</td></tr><tr><td>4</td><td>Lock page 7</td></tr><tr><td>5</td><td>Lock page 8</td></tr><tr><td>6</td><td>Lock page 9</td></tr><tr><td>7</td><td>Lock page 10</td></tr><tr><td>8</td><td>Lock page 11</td></tr><tr><td>9</td><td>Lock page 12</td></tr><tr><td>10</td><td>Lock page 13</td></tr><tr><td>11</td><td>Lock page 14</td></tr><tr><td>12</td><td>Lock page 15</td></tr><tr><td>13</td><td>Lock page 16-19</td></tr><tr><td>14</td><td>Lock page 20-23</td></tr><tr><td>15</td><td>Lock page 24-27</td></tr><tr><td>16</td><td>Lock page 28-31</td></tr><tr><td>17</td><td>Lock page 32-35</td></tr><tr><td>18</td><td>Lock page 36-39</td></tr><tr><td>19</td><td>Lock page 41</td></tr><tr><td>20</td><td>Lock page 42</td></tr><tr><td>21</td><td>Lock page 43</td></tr><tr><td>22</td><td>Lock page44-47 (Key Area)</td></tr></table>	Value	Description	0	Lock page 3 (OTP Area)	1	Lock page 4	2	Lock page 5	3	Lock page 6	4	Lock page 7	5	Lock page 8	6	Lock page 9	7	Lock page 10	8	Lock page 11	9	Lock page 12	10	Lock page 13	11	Lock page 14	12	Lock page 15	13	Lock page 16-19	14	Lock page 20-23	15	Lock page 24-27	16	Lock page 28-31	17	Lock page 32-35	18	Lock page 36-39	19	Lock page 41	20	Lock page 42	21	Lock page 43	22	Lock page44-47 (Key Area)
				Value	Description																																															
				0	Lock page 3 (OTP Area)																																															
				1	Lock page 4																																															
				2	Lock page 5																																															
				3	Lock page 6																																															
				4	Lock page 7																																															
				5	Lock page 8																																															
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				15	Lock page 24-27																																															
				16	Lock page 28-31																																															
17				Lock page 32-35																																																
18				Lock page 36-39																																																
19	Lock page 41																																																			
20	Lock page 42																																																			
21	Lock page 43																																																			
22	Lock page44-47 (Key Area)																																																			

| Return | 0: Success Not 0: Failure, please check Error code table. | | | |
| Reference code | | | | |

3.8.5.ULTRALIGHT_LockBlock 【Lock Block】

Definition	err_t ULTRALIGHT_LockBlock(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE blockType)					
Description	Some bits of the lock byte of the page 2 or page 40, once these bits are locked, they can not be changed.					
Parameters	Name	Type	Direction	Remarks		
	hr	RFID_READER_HANDLE	[in]	Reader handle		
	ht	RFID_TAG_HANDLE	[in]	The tag handle created by ULTRALIGHT_connect		
	blockType	BYTE	[in]	Value	Description	
				0	Lock byte OTP bit	
				1	Lock byte page 4-9 bit	
				2	The lock bit of the lock byte page 10-15	
				3	The lock bit of the lock byte page 16-27	
				4	The lock bit of the lock byte page 28-39	
				5	Count Lock bit of the lock byte	
				6	Auth0 Lock bit of the lock byte	
				7	Auth0 Lock bit of the lock byte	
8	The lock bit of the key area of the lock byte					
Return	0: Success Not 0: Failure, please check Error code table.					
Reference code						

3.8.6. ULTRALIGHT_ReadMultiplePages 【Read multiple pages】

Definition	err_t ULTRALIGHT_ReadMultiplePages(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE pageStart, BYTE pageNum , BYTE databuf[] ,DWORD *nSize)			
Description	Read multiple pages			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle created by ULTRALIGHT_connect .
	pageStart	BYTE	[in]	Start page , ultralight page address 0-15; ultralight C page address 0-47;
	pageNum	BYTE	[in]	Pages amount
	databuf	BYTE[]	[out]	Read the page data
	nSize	DWORD *	[in,out]	In: The maximum number of bytes that can be written to the databuf buffer; Output: The number of bytes written to the databuf buffer;
Return	0: Success Not 0: Failure , please check Error code table.			
Reference code				

3.8.7. ULTRALIGHT_WriteMultiplePages 【Read multiple pages】

Definition	err_t ULTRALIGHT_WriteMultiplePages(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE pageStart,BYTE pageNum , BYTE databuf[] ,DWORD bytesToWrite)			
Description	Write multiple pages			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle created by ULTRALIGHT_connect
	pageStart	BYTE	[in]	Page start, ultralight page address 0-15; ultralight C page address 0-47;
	pageNum	BYTE	[in]	Pages amount
	databuf	BYTE[]	[in]	The page data to be written
	nSize	DWORD *	[in]	The number of bytes to write
Return	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.8.8. ULTRALIGHT_Authenticate 【Authentication】

Definition	err_t ULTRALIGHT_Authenticate(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE key[] /* 16bytes */)			
Description	Key authentication, the API only available with Ultralight C. If tag protected by password, it must be authenticated before reading or writing data pages.			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle created by ULTRALIGHT_connect
	key	BYTE []	[in]	Secret key of 16 byte
Return	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.8.9. ULTRALIGHT_UpdatePassword 【Update the key】

Definition	err_t ULTRALIGHT_UpdatePassword(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE key[])			
Description	Update key,the API only available for Ultralight C. If the tag protected by password, it must be authenticated before using this function, and there can't be other read or write operations.			
Parameters	Name	Type	Directions	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ULTRALIGHT_connect
	key	BYTE []	[in]	New secret key of 16 byte
Return	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.8.10. ULTRALIGHT_UpdateAuthConfig 【Update the key configuration】

Definition	err_t ULTRALIGHT_UpdateAuthConfig(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE auth0, BYTE auth1)					
Description	Update key,the API only available for Ultralight C. If the tag is password protected, it must be authenticated before using this function, and there can't be other read or write operations.					
Parameters	Name	Type	Direction	Remarks		
	hr	RFID_READER_HANDLE	[in]	Reader handle		
	ht	RFID_TAG_HANDLE	[in]	Tag handle created by ULTRALIGHT_connect		
	auth0	BYTE	[in]	Authentication Configuration Byte 0, sets the start page to be protected 0x03-0x30 .		
	auth1	BYTE	[in]	Authentication Configuration Byte 1		
				Value	Description	
				bit0	0	Read and write require key authentication
					1	Write requires key authentication, read no required
bit1-7	keep					
Return	0: Success Not 0: Failure , please check Error code table.					
Reference code						

3.8.11. ULTRALIGHT_AuthenSafe 【Safe Authentication】

Definition	err_t ULTRALIGHT_AuthenSafe(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht)			
Description	Safe Authentication, need to combine RDR_LoadAuthKey with RDR_SelectAuthKey			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	The handle of the tag created by ULTRALIGHT_connect
Return	0: Success Not 0: Failure, please check Error code table.			
Reference code				

3.9.NTAG21X Function Interface

3.9.1.NTAG21x_Connect

Definition	err_t RFIDLIB_API NTAG21x_Connect(RFID_READER_HANDLE hr,BYTE uid[] /* 7 bytes */ ,RFID_TAG_HANDLE *ht) ;			
Instruction				
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	uid	BYTE[]	[in]	Tag UID
	ht	RFID_TAG_HANDLE	[out]	Tag handle, created by NTAG21x_Connect.
Return	0: Success Not 0: Failure, please check Error code table.			
Reference Code				

3.9.2.NTAG21x_GetVersion

Definition	err_t RFIDLIB_API NTAG21x_GetVersion(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE ver[] /* out */);			
Instruction				
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by NTAG21x_Connect.
	ver	BYTE 【8】	[out]	Get product version information for the NTag21x tag
Return	0: Success Not 0: Failure, please check Error code table.			
Reference Code				

3.9.3.NTAG21x_Authenticate

Definition	err_t RFIDLIB_API NTAG21x_Authenticate(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, DWORD pwd /* in */, WORD *pack /*out */)			
Instruction				
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by NTAG21x_Connect.
	pwd	DWORD	[in]	Secret Key
	pack	WORD*	[out]	Key password confirmation
Return	0: Success Not 0: Failure, please check Error code table.			
Reference Code				

3.9.4.NTAG21x_ReadMultiplePages

Definition	err_t RFIDLIB_API NTAG21x_ReadMultiplePages(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, DWORD pageStart, DWORD pageNum , BYTE databuf[] , DWORD *nSize) ;			
Instruction				
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by NTAG21x_Connect.
	pageStart	DWORD	[in]	Start page
	pageNum	DWORD	[in]	The number of pages read
	databuf	BYTE[4*pageNum]	[out]	The data has been read
	nSize	DWORD *	[in,out]	Indicates the length of the databuf array passed in to the function, and returns the actual obtained data length
Return	0: Success Not 0: Failure, please check Error code table.			
Reference Code				

3.9.5.NTAG21x_WriteMultiplePages

Definition	err_t RFIDLIB_API NTAG21x_WriteMultiplePages(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, DWORD pageStart, DWORD pageNum , BYTE databuf[] , DWORD bytesToWrite) ;			
Instruction				
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by NTAG21x_Connect.
	pageStart	DWORD	[in]	Start page for writing data
	pageNum	DWORD	[in]	The number of pages written
	databuf	BYTE[4*pageNum]	[out]	Data to be written
	bytesToWrite	DWORD	[in]	Indicates the length of the databuf array passed in to the function
Return	0: Success Not 0: Failure, please check Error code table.			
Reference Code				

3.9.6.NTAG21x_ReadSig

Definition	err_t RFIDLIB_API NTAG21x_ReadSig(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE sig[] /* out */);			
Instruction				
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by NTAG21x_Connect.
	sig	BYTE 【32】	[out]	Obtain the signature of the NTag21x tag
Return	0: Success Not 0: Failure, please check Error code table.			
Reference Code				

Appendix 1.MIFARE® Ultralight Memory structure

Page address	Byte 0	Byte 1	Byte 2	Byte 3	
0	Serial number	Serial number	Serial number	Serial number	
1	Serial number	Serial number	Serial number	Serial number	
2	Serial number	Internal	Lock bit	Lock bit	
3	OTP	OTP	OTP	OTP	
4	User area	User area	User area	User area	
5	User area	User area	User Area	User area	
6	User area	User area	User Area	User area	
7	User area	User area	User Area	User area	
8	User area	User area	User Area	User area	
9	User area	User area	User Area	User area	
.....	User area	User area	User area	User area	
15	User area	User area	User area	User area	

Appendix 2.Data block access conditions of MIFARE® classic

Access Type	Read	Write	Increase	decrease
0	key A B	key A B	key A B	key A B
2	key A B	never	never	never
1	key A B	key B	never	never
3	key A B	key B	key B	key A B
4	key A B	never	never	key A B
6	key B	key B	never	never
5	key B	never	never	never
7	never	never	never	never

Appendix 3.Tail block access conditions of MIFARE® classic

Access Type	Key A		Access conditions		Key B	
	Read	Write	Read	Write	Read	Write
0	never	key A	key A	never	key A	key A
2	never	never	key A	never	key A	never
1	never	key B	key A B	never	never	key B
3	never	never	key A B	never	never	never
4	never	key A	key A	key A	key A	key A
6	never	key B	key A B	key B	never	key B
5	never	never	key A B	key B	never	never
7	never	never	key A B	never	never	never